



LECTURE 6

IMPLICIT NEURAL REPRESENTATIONS

Representing geospatial information as a continuous
learned function

PRESS – OR SPACE. →

Learning Outcomes

- 1 Explain how geospatial information can be represented as **discrete data or continuous functions**.
- 2 Explain how **coordinate-based neural networks** model continuous signals using positional encodings and sinusoidal features.
- 3 Explain how **implicit neural representations** can model geographic space and produce either spatial signals or location embeddings.

Where We Are in the Course

LECTURE 4

Model & Optimization

How does a neural network learn from labeled data?

LECTURE 5

Generalization

How does learned structure transfer beyond the training data?

LECTURE 6 — TODAY

Representing Data as Functions

Can geospatial information be represented as a continuous learned function instead of discrete samples?

Main question: Can geospatial information be represented as a continuous learned function instead of only as discrete samples?

Discrete or Continuous?

DISCRETE

Store observations explicitly at discrete spatial supports.

$$\{(\mathbf{x}_i, y_i)\}_{i=1}^N$$

Examples: raster cells · vector geometries · point samples

CONTINUOUS

Store the parameters of a function; evaluate wherever needed.

$$f : \mathcal{X} \rightarrow \mathcal{Y}$$

Examples: Fourier representations · spherical harmonic models · interpolation functions · **neural fields / implicit neural representations**

Neural fields are one type of continuous functional model — not a separate category.

Geospatial analysis commonly discretizes space — but global physical fields have always been modeled as continuous functions.

Functional Models of the Earth

The Earth as Discrete Data

RASTER

Regular grid of cells, each holding a value

VECTOR

Points, lines, polygons with explicit geometry

POINT OBSERVATIONS

Irregular samples at measured locations

Most geospatial tools discretize: store values at explicit supports, then process those values.

What Happens at Global Scale?

Some geophysical quantities vary continuously over the entire Earth:

- Gravity field
- Magnetic field
- Geoid (sea-level surface)
- Atmospheric pressure
- Temperature and precipitation

[global field image]

Must a global field be stored as billions of individual pixels?

From Samples to Functions

General functional model:

$$f(x) = \sum_k a_k \phi_k(x)$$

- ϕ_k : basis functions — predefined shapes that span the space
- a_k : coefficients — the learned or fitted parameters
- The coefficients *are* the stored representation of the field

DISCRETE

Store N values at N locations

FUNCTIONAL

Store K coefficients; evaluate $f(x)$ anywhere

Sine and Cosine as Basis Functions

The simplest basis functions on a periodic domain are sinusoids:

$$\phi_k(x) = \sin(kx), \quad \cos(kx)$$

- Each basis function oscillates at a different frequency k
- Together they can represent any square-integrable periodic signal

[sinusoidal basis functions at $k = 1, 2, 3, \dots$]

This prepares us for Fourier representations and, later, positional encoding in neural networks.

Fourier Representations

Any signal can be decomposed into components at different spatial frequencies:

$$f(x) = \sum_k (a_k \cos(kx) + b_k \sin(kx))$$

Low frequencies — broad, smooth variation

High frequencies — fine spatial detail

[Fourier decomposition figure]

→ This connects directly to **spectral bias** and **positional encoding** in Block 2.

From the Circle to the Sphere

Fourier functions work on periodic or Euclidean domains. The Earth is a sphere.

$$\begin{aligned} & \sin, \cos \\ & \quad \rightarrow \text{sphere} \\ & Y_{\ell m}(\theta, \lambda) \end{aligned}$$

Spherical harmonics are the natural generalization of sinusoidal basis functions to the sphere S^2 .

[circle → sphere transition figure]

Spherical Harmonics

A field on the sphere is modeled as:

$$f(\theta, \lambda) = \sum_{\ell, m} a_{\ell m} Y_{\ell m}(\theta, \lambda)$$

- ℓ : degree — controls the spatial scale
- m : order — controls the longitudinal variation
- Increasing ℓ adds finer spatial detail

[spherical harmonic visualization $Y_{\ell m}$]

Modeling Global Earth Fields

Spherical harmonics are the standard representation in physical geodesy:

- Gravity field (e.g. EGM2008: degree 2190)
- Geomagnetic field (IGRF model)
- Earth's topography

The field is stored through coefficients of a continuous functional model — not as billions of pixels.

[global field represented via spherical harmonics]

Where Do the Basis Functions Come From?

So far we chose basis functions by hand (sinusoids, spherical harmonics).

Can the **geometry of the domain itself** determine useful basis functions?

The answer is yes — through the eigenfunctions of the Laplacian.

The Laplacian

The Laplacian Δ measures local variation of a function on a domain.

- On $\mathbb{R}^n$: $\Delta f = \sum_i \partial^2 f / \partial x_i^2$
- Captures local smoothness
- Generalizes naturally to manifolds and graphs
- Encodes the geometry of the domain

[Laplacian as local variation]

Eigenfunctions of the Laplacian

Eigenfunctions ϕ_k satisfy:

$$\Delta \phi_k = \lambda_k \phi_k$$

- λ_k : eigenvalue — controls the spatial frequency
- ϕ_k : eigenfunction — a basis shape adapted to the domain
- Together they form a geometry-aware basis for representing any function on the domain

[Laplacian eigenfunctions on a domain]

Geometry Determines the Basis

LINE / PERIODIC

$$\sin(kx), \cos(kx)$$

Fourier modes

SPHERE S^2

$$Y_{\ell m}(\theta, \lambda)$$

Spherical harmonics

GENERAL MANIFOLD / GRAPH

$$\Delta \phi_k = \lambda_k \phi_k$$

Laplacian eigenfunctions

Each domain determines its own natural frequency basis through the geometry.

Block 1 Takeaway

A spatial field can be represented by the *parameters of a continuous function* rather than by explicitly storing every sample.

Classical geodesy and signal processing have exploited this for decades: spherical harmonic models of gravity, magnetic, and other global fields store coefficients, not pixels.

Next: what if the continuous function is parameterized by a neural network?

Fixed functional models choose basis functions by hand. Neural networks learn the function directly from data.

Neural Networks for Continuous Signal Modeling

From Fixed Functional Models to Neural Networks

CLASSICAL: FIXED BASIS

$$f(x) = \sum_k a_k \phi_k(x)$$

Basis functions ϕ_k are fixed; only coefficients a_k are learned.

NEURAL: LEARNED FUNCTION

$$f_{\theta}(x) = \mathrm{NN}_{\theta}(x)$$

The entire function is parameterized by network weights θ .

What if the continuous function itself is parameterized by a neural network?

Coordinate-Based Neural Networks

The most direct formulation: the input is a coordinate; the output is the signal at that coordinate.



An Image as a Continuous Function

DISCRETE: PIXEL GRID

[pixel grid]

Store $H \times W \times 3$ values explicitly

CONTINUOUS: NEURAL FUNCTION

$$f_{\theta}(x, y) \rightarrow \mathrm{RGB}$$

Store network weights θ ; evaluate at any (x,y)

Fit the Samples, Query the Function

TRAINING

$$\min_{\theta} \sum_i \|f_{\theta}(\mathbf{x}_i) - y_i\|^2$$

Observed coordinates and values supervise the network.

INFERENCE

$$y^* = f_{\theta}(\mathbf{x}^*)$$

Evaluate the function at any arbitrary coordinate.

The network learns a continuous function from discrete observations — much like classical interpolation, but with a learned parametric form.

Spectral Bias

Standard MLPs with ReLU activations preferentially fit **low-frequency** variation first.

- Smooth, broad patterns are learned quickly
- Fine high-frequency structure is learned slowly or not at all

Recall from Block 1: Fourier decomposition separates signals by frequency. Spectral bias is a neural analogue of truncating the Fourier series.

[spectral bias: MLP learns low frequencies first]

Positional Encoding and Fourier Features

Idea: map coordinates to explicit sinusoidal features before the MLP.

$$\gamma(\mathbf{x}) = \text{bigl}[\sin(2\pi B\mathbf{x}), \cos(2\pi B\mathbf{x})\bigr]$$

followed by an MLP on $\gamma(\mathbf{x})$.

- B : frequency matrix (random or learned)
- Directly injects multiple frequency scales
- Overcomes spectral bias for fine detail

[Fourier feature encoding diagram]

Tancik et al. 2020

Why Fourier Features Help

RAW COORDINATES

$$(x, y) \rightarrow \mathrm{MLP}$$

MLP must learn all frequency content from scratch. Spectral bias limits high-frequency capacity.

FOURIER-ENCODED COORDINATES

$$\gamma(x, y) \rightarrow \mathrm{MLP}$$

Multiple frequencies are already explicit in the input. MLP only needs to combine them.

Explicit frequency features make fine spatial variation much easier for the network to represent.

SIREN

Alternative approach: use sinusoidal *activation functions* inside the network.

$$h_{l+1} = \sin(W_l h_l + b_l)$$

Key distinction:

Fourier features — sinusoidal *input encoding*

SIREN — sinusoidal *activation functions inside the network*

[SIREN network diagram]

Sitzmann et al. 2020

Fourier Basis, Fourier Features, SIREN

CLASSICAL SIGNAL PROCESSING

$$f(x) = \sum_k a_k \phi_k(x)$$

Fourier basis: fixed sinusoidal functions

NEURAL INPUT ENCODING

$$\gamma(\mathbf{x}) = [\sin(B\mathbf{x}), \cos(B\mathbf{x})]$$

Fourier features: sinusoidal input to MLP

SIREN ACTIVATIONS

$$h_{l+1} = \sin(W_l h_l + b_l)$$

Sinusoidal activations inside the network

The same idea — spatial frequency decomposition — appears at every level of the representation.

Continuous Neural Fields

The coordinate-network idea generalizes across many signal types:

Images

$(x, y) \rightarrow \mathrm{RGB}$

Audio

$t \rightarrow \text{amplitude}$

Signed-distance functions

$(x, y, z) \rightarrow \mathrm{SDF}$

Occupancy fields

$(x, y, z) \rightarrow \{0, 1\}$

Physical fields

$\mathbf{x} \rightarrow \text{temperature}$

Radiance fields (NeRF)

$(\mathbf{x}, \mathbf{d}) \rightarrow (\sigma, \mathrm{RGB})$

NeRF

NeRF (Neural Radiance Field) models a 3D scene as a continuous function:

$$(\mathbf{x}, \mathbf{d}) \rightarrow (\sigma, \mathbf{c})$$

- $\mathbf{x}$: 3D position
- $\mathbf{d}$: viewing direction
- σ : volume density
- $\mathbf{c}$: RGB color

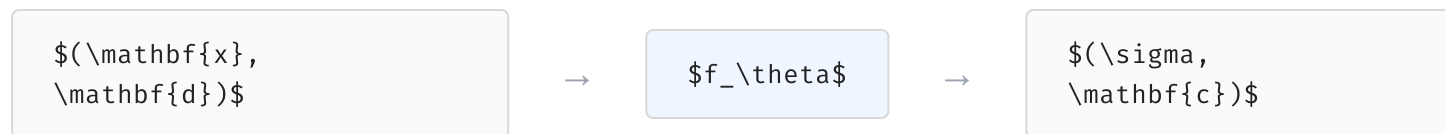
Novel views are rendered by ray-marching through the neural field.

[NeRF scene representation]

Mildenhall et al. 2020

A Scene as a Continuous Neural Field

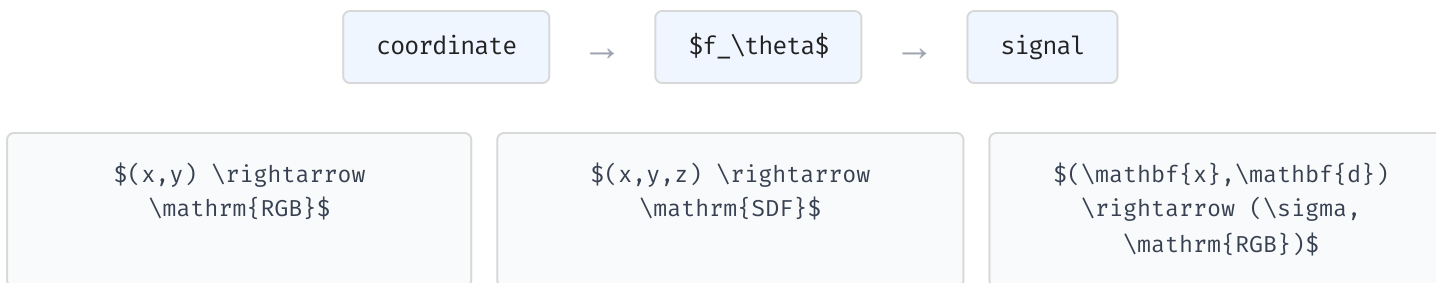
The conceptual key: a 3D scene is not stored as explicit geometry or a point cloud — it is the network itself.



Any point in the scene can be queried through the neural function — no explicit mesh, voxel grid, or point set required.

The Same Pattern Across Domains

Common pattern:



Block 2 Takeaway

Neural networks can parameterize continuous functions and model signals directly from coordinates.

Spectral bias limits plain MLPs. Positional encodings (Fourier features) and sinusoidal activations (SIREN) overcome this by explicitly representing spatial frequencies — echoing Block 1's functional models.

Next: what changes when the domain is the Earth?

Synthesizing Blocks 1 and 2: geometry-aware coordinate encodings meet neural continuous functions.

Implicit Neural Representations of Geographic Space

What Changes When the Domain Is the Earth?

EUCLIDEAN IMAGE PLANE

- Flat, periodic domain
- Uniform distance metric
- (x, y) coordinates well-defined
- Fourier features work directly

THE SPHERE

- Longitude is discontinuous ($-180^\circ/+180^\circ$)
- Poles have degenerate coordinates
- Spherical geometry, not Euclidean
- Distance in lat/lon is non-uniform

Coordinate encodings designed for flat space may fail or distort when applied naively to geographic coordinates.

Geographic Coordinates Are Not Just x and y

LATITUDE / LONGITUDE

Simple but discontinuous; non-uniform distances

3D CARTESIAN ON SPHERE

Continuous and uniform; constrains $x^2+y^2+z^2=1$

FOURIER FEATURES

Fast; may distort at poles and date line

SPHERICAL HARMONICS / LAPLACIAN

Geometry-aware; respects spherical topology

The choice of coordinate encoding determines how well the model respects the geometry of geographic space.

Geometry-Aware Coordinate Encodings

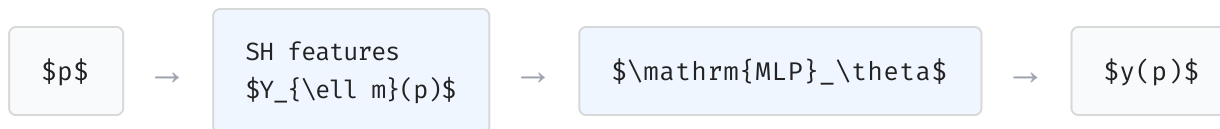
Reconnecting to Block 1: spherical harmonics can serve not only as a complete field model, but also as a **coordinate encoding** for a neural network.

$$p \mapsto Y_{\ell m}(p)$$

Map geographic location $p = (\theta, \lambda)$ to a vector of spherical harmonic values as features.

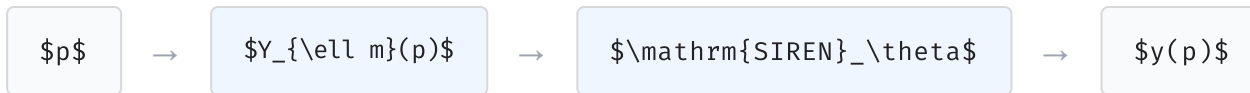
[geographic coord → SH encoding → MLP]

Spherical Harmonics + Neural Network



The spherical harmonic encoding provides a **geometry-aware** positional representation that respects spherical topology before the neural network processes it.

Spherical Harmonics + SIREN



Spherical harmonics

Geometry-aware positional features; respect spherical topology

SIREN

Nonlinear high-frequency modeling capacity beyond the SH basis

Residual Implicit Networks

Residual coordinate network variants (e.g. ResIREN) extend the basic SIREN with skip connections:



One Global Network or Many Local Networks?

ONE GLOBAL MLP

Single network for the entire Earth. Simple but may lack local resolution.

MULTIPLE LOCAL MLPs

One network per tile or region. High resolution; costly to scale.

HIERARCHICAL / MULTI-SCALE

Coarse global + fine local network. Balanced capacity and resolution.

MIXTURE-OF-EXPERTS

Dynamic routing to specialized sub-networks per location or scale.

Scaling continuous neural fields to large geographic domains is an active research direction.

Geographic Neural Fields

General supervised formulation:

$$(p_i, y_i) \rightarrow f_{\theta}(p) \rightarrow y$$

Possible output signals:

- Climate variables
- Elevation
- Land cover / land use
- Population density
- Ecological observations

[predicted geographic signal map]

Species Observations as a Spatial Signal

Biodiversity observations (e.g. iNaturalist) provide sparse, presence-only spatial data:

$$\text{location} \rightarrow P(\text{species} \mid \text{location})$$

A geographic neural field can estimate species distribution continuously from irregular observations.

[species observation points → predicted distribution]

From Signal Prediction to Location Representation

SIGNAL MODEL

$$f_{\theta}(p) \rightarrow y$$

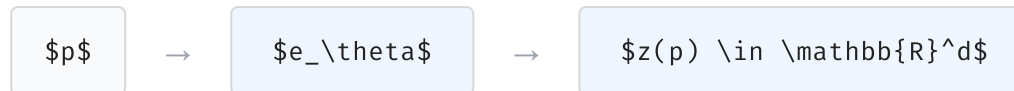
Predict one specific output (temperature, species, elevation) at any location.

LOCATION ENCODER

$$p \mapsto e_{\theta}; z(p) \rightarrow y$$

Learn a *reusable* representation $z(p)$ that can support many downstream tasks.

Location Encoders

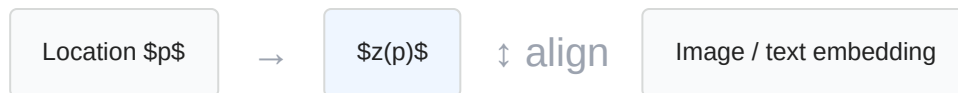


The output is a **reusable representation** of geographic location, rather than a prediction for one particular signal.

This is analogous to how image encoders learn general visual features rather than predicting a fixed label. The downstream task uses $z(p)$ as input rather than the raw coordinate.

GeoCLIP-Like Location Representations

Multimodal and contrastive learning approaches train location encoders by aligning geographic coordinates with other modalities (images, text, remote sensing).



Contrastive learning details are covered in a later lecture. Here the key point is that the location encoder output is a general-purpose geographic embedding.

Signal Model or Representation Model?

IMPLICIT SIGNAL MODEL

$$p \rightarrow y$$

One network, one output type. Train and use for a specific signal.

LOCATION ENCODER

$$p \rightarrow z$$

Reusable embedding of location.
Downstream task uses z .

DOWNSTREAM TASK

$$z \rightarrow y$$

Lightweight head applied to the fixed location representation.

Design Space of Geographic INRs

DOMAIN

plane · sphere · manifold

COORDINATE ENCODING

raw lat/lon · Fourier · spherical
harmonics · Laplacian

NEURAL MODEL

MLP · SIREN · residual network

SCALE

global · local · hierarchical

OUTPUT

signal · probability · embedding

TRAINING

supervised · multimodal · self-
supervised

Neural Fields of the Earth

Earth coordinate → geometry-aware encoding → neural function →

geospatial signal
– or –
location embedding

Lecture Synthesis

BLOCK 1

Functional Models of the Earth

$$p \mapsto \phi(p) \\ \mapsto y$$

Fixed basis; fitted coefficients

BLOCK 2

Neural Networks for Continuous Signals

$$p \mapsto \gamma(p) \\ \mapsto f_{\theta} \\ \mapsto y$$

Learned encoding; learned function

BLOCK 3

Implicit Neural Representations of Geographic Space

$$p \mapsto \text{geo encoding} \\ \mapsto f_{\theta} \\ \mapsto \text{or } z$$

Geometry-aware; signal or embedding

Geospatial information can be stored **explicitly** at discrete locations, or **implicitly** through the parameters of a continuous function that can be queried throughout space.

Location Encoders & Spatial Embeddings

Representing space continuously

In this lecture (original outline)

- Coordinates as model inputs
- Positional and coordinate encodings
- Location encoders
- Spatial embeddings
- Implicit neural representations and continuous spatial functions